

Deck content, rewritten against what is actually built

Same six-slide template, same section order. The copy below replaces the current text so the submission matches the working system — and adds the architecture, the Google Cloud recognition path, and screenshots of the running dashboard.

How to use this. Each section maps to one slide. Copy the text into the template as-is; the block diagrams and the dashboard screenshots are sized for the slide area. Full-resolution images are saved in docs/slides/ — insert those into PowerPoint rather than screenshotting this page.

Two labels appear throughout. **BUILT** means demonstrable today. **PLANNED** means designed and scoped, not yet running. Keep them honest in the deck; a judge who asks for a demo of a "Planned" item and gets one is a judge you have lost.

01 Title slide

unchanged, except the title line

- **Problem Statement ID** — 26034
- **Problem Statement Title** — Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels.
- **Theme** — Miscellaneous · **PS Category** — Software
- **Team Name** — Blitz

Give the idea a name. The template has an "IDEA TITLE" field that is currently empty. Suggested: **"Cited & Checkable — a compliance finding an officer can defend"**. It states the differentiator in four words, which is the whole pitch.

PROPOSED SOLUTION

- **Point, capture, cited verdict.** An officer photographs the declaration panel; the app locates each mandatory declaration and returns a verdict in seconds, on the device, with no network. **BUILT**
- **Every flag cites its clause.** Statute, rule, sub-clause and rule-pack version travel with each finding — and where a sub-clause letter is itself disputed, both readings are shown rather than one being chosen silently. **BUILT**
- **The law is data, not code.** No clause number, threshold or lexicon appears in any source file; all of it lives in versioned JSON rule packs carrying per-clause review status. A change in the Rules is a data edit, not a software release. **BUILT**
- **It refuses rather than guesses.** Frame quality is measured against thresholds held in the pack; below them no verdict is offered and the failed check is named. A poor photograph never becomes a false accusation against a manufacturer. **BUILT**
- **A repository for controllers.** Web dashboard showing which declaration is most often missing across a district, repeat offenders by manufacturer, and a governance screen listing every clause and its verification state. **BUILT**
- **Next:** Unit Sale Price validation with Rule 6(11) exemptions, Rule 7 letter-height auditing against a scale reference, GS1/GEPIR identity cross-check, and batch sweeps over e-commerce listings.

PLANNED

Perception is swappable. Judgement is not.

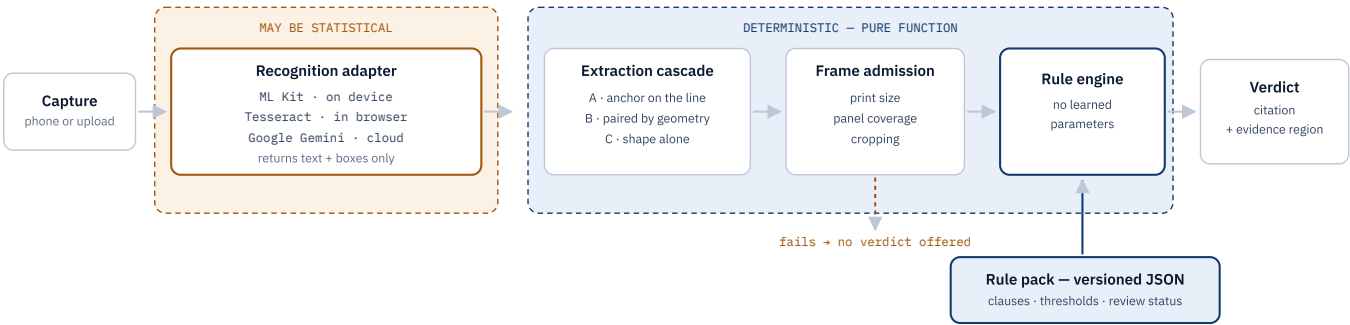
The recogniser sits behind an interface — ML Kit on the phone, Tesseract in the browser, Google Gemini in the cloud. The step that turns text into a legal finding is a pure function with no learned parameters. We proved the seam by replacing the engine three times without changing anything downstream.

Evidence an officer can argue with

Every finding carries the image region behind it, and every report prints the raw recognised text. When the machine misreads, that is visible — not hidden behind a confident verdict. This is what makes a finding survive a manufacturer's challenge.

Governance built into the artefact

Each rule-pack entry carries a review status. While unreviewed, every finding is capped at *advisory* — enforced by a test, not by convention — so nothing can be presented as a determination of law before a Legal Metrology officer signs it off.



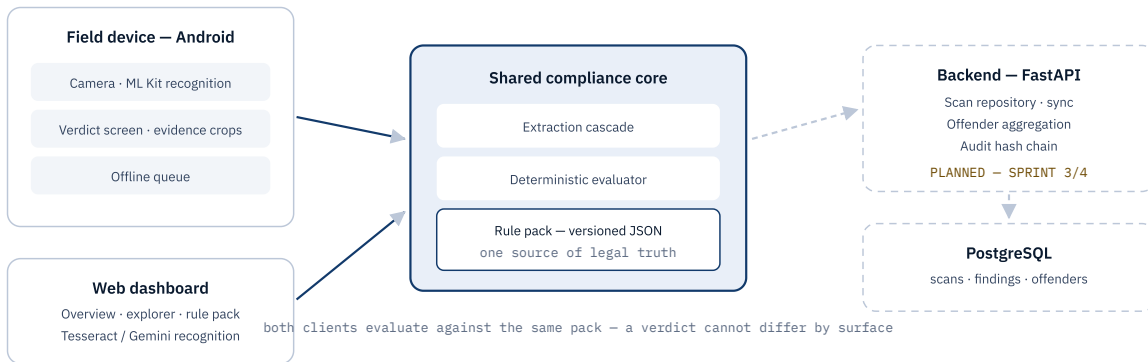
The boundary is the point: everything left of it may be statistical, everything right of it is a pure function of (fields, geometry, rule pack).

TECHNOLOGY STACK

LAYER	CHOICE
Mobile	React Native + Expo dev-client (Android), ML Kit text recognition BUILT
Web	React 19 + Vite + Tailwind, hand-rolled SVG charts BUILT
Recognition	OcrProvider interface — ML Kit (device) · Tesseract WASM (browser, offline) · Google Gemini via AI Studio (cloud) BUILT · Cloud Vision PLANNED
Rule engine	TypeScript pure function, shared by app and dashboard; rule packs as versioned JSON BUILT
Geometry	Anchor-to-value association in multiples of text height — scale-free, no pixel constants BUILT
Backend	FastAPI + PostgreSQL, JWT role-based access PLANNED
Reports	Printable inspection note carrying every citation BUILT · WeasyPrint PDF service PLANNED
Quality	158 automated tests; conformance fixtures both evaluators must agree on BUILT

THE GOOGLE CLOUD RECOGNITION PATH

- **What it does.** Transcribes the label and returns line boxes. It is never asked whether anything is compliant — the prompt requests transcription and geometry only.
- **Why that boundary matters.** A model's opinion is not reproducible; an enforcement action must be. Keeping the model on the perception side preserves the determinism claim entirely.
- **Why it is needed.** On creased, glossy Indian packaging, on-device OCR degrades badly. A vision model resolves a damaged glyph from context, the way a person does.
- **Offline is preserved.** Tesseract remains the default and the fallback; the Android app reaches a verdict with no network at all. Cloud is an enhancement path, never a dependency.
- **Production route.** Because it is an adapter, moving to a self-hosted model or a vendor under a data-processing agreement is a configuration change, not a rewrite.



Dashed boundaries are scoped and specified, not yet running.

FEASIBILITY & VIABILITY

- **Hardware.** A standard Android phone. No depth sensor, no reference card, no attachment — the geometry is expressed in multiples of the label's own text height, so it scales with camera distance instead of depending on it.
- **Offline by construction.** Recognition, rule pack and verdict all run on the device. The web build ships its recognition models too, so it works with the network unplugged.
- **Operational fit.** An officer checks roughly eighty packages a day. The bottleneck is not spotting text — it is producing a finding that survives challenge. That is what the tool automates.
- **Amendment cost.** A change in the Rules is a JSON edit with a version and a reviewer, not a software release cycle across a fleet of devices.
- **Cost.** No GPU, no training data, no per-device licence. Cloud recognition is optional and, at inspection volumes, effectively free.

CHALLENGES & MITIGATIONS

1 · Glare, creases and curved packaging

Measured, not assumed. On a creased foil packet, on-device OCR read "Manufactured by" as sanedoched by. Mitigation: recognition is an adapter, so a stronger engine is a configuration change; the image is read several ways and the reading that survives its own quality checks is kept; and when nothing is good enough the tool refuses rather than guesses.

2 · Multilingual labels

Devanagari is read on-device by ML Kit and by the cloud model. Anchors and lexicons are pack data, so a language is added as data. Where a script is unsupported the app says so rather than silently returning less.

3 · Confidently wrong readings — the real risk

A missed declaration costs a rescan; a fabricated one costs an officer's credibility. Every report prints the raw recognised text beside the verdict, every finding carries its image region, and unreviewed clauses cannot emit anything stronger than *advisory*.

Legal Metrology
Compliance Checker

Overview

New scan

Scan explorer

Rule pack

How it works

Compliance report

Legal Metrology (Packaged Commodities) Rules, 2011

Sample repository

Advisory only. The active rule pack is pending legal review — every finding is capped at advisory and none may be presented as a determination of law.

SC-24875

Needs attention

sample record

All scans

Download notice

Surf Excel — Detergent powder

At least one declaration did not pass. Each finding cites the rule it comes from.

CAPTURED

10 Sept 2026, 11:30 pm

OFFICER

M. Sheikh

DISTRICT

Chhatrapati Sambhajinagar

CATEGORY

Household cleaning

SOURCE

Field device

RULE PACK

demo-lmpc-v0 v0.1.0

READ BY

sample record — not read fr...

Evidence

Drawn from the sample record — not a photograph

SURF EXCEL

Detergent powder

DECLARATIONS

Net Qty: 500

M.R.P. ₹99.00 (incl. of all taxes)

Mfg. / Packed: 07/2026

MANUFACTURED / PACKED BY
Mfd. by Hindustan Unilever Ltd., Mumbai 400099, Maharashtra

CONSUMER CARE
care@example.in · 1800 220 3000

ILLUSTRATIVE — DRAWN FROM THE SAMPLE RECORD, NOT A PHOTOGRAPH

Declarations read from the label

6 of 6 located

Manufacturer / packer / importer name and address

Mfd. by Hindustan Unilever Ltd., Mumbai 400099, Maharashtra

Read from the anchor's own line

high

Common or generic name of the commodity

Detergent powder

Read from the anchor's own line

high

Net quantity

500

Paired with its label by geometry — the value sits 1.2 text-heights to the right of the anchor, on the same row

high

Retail sale price (MRP), inclusive of all taxes

₹99.00 (incl. of all taxes)

Read from the anchor's own line

medium

Month and year of manufacture, packing or import

07/2026

Paired with its label by geometry — the value sits 1.5 text-heights to the right of the anchor, on the same row

high

Consumer care contact

care@example.in · 1800 220 3000

Read from the anchor's own line

medium

Rule checks (1 of 8 flagged)

Every check the pack defines, and how this package answered it

Rule 6(1)(a)

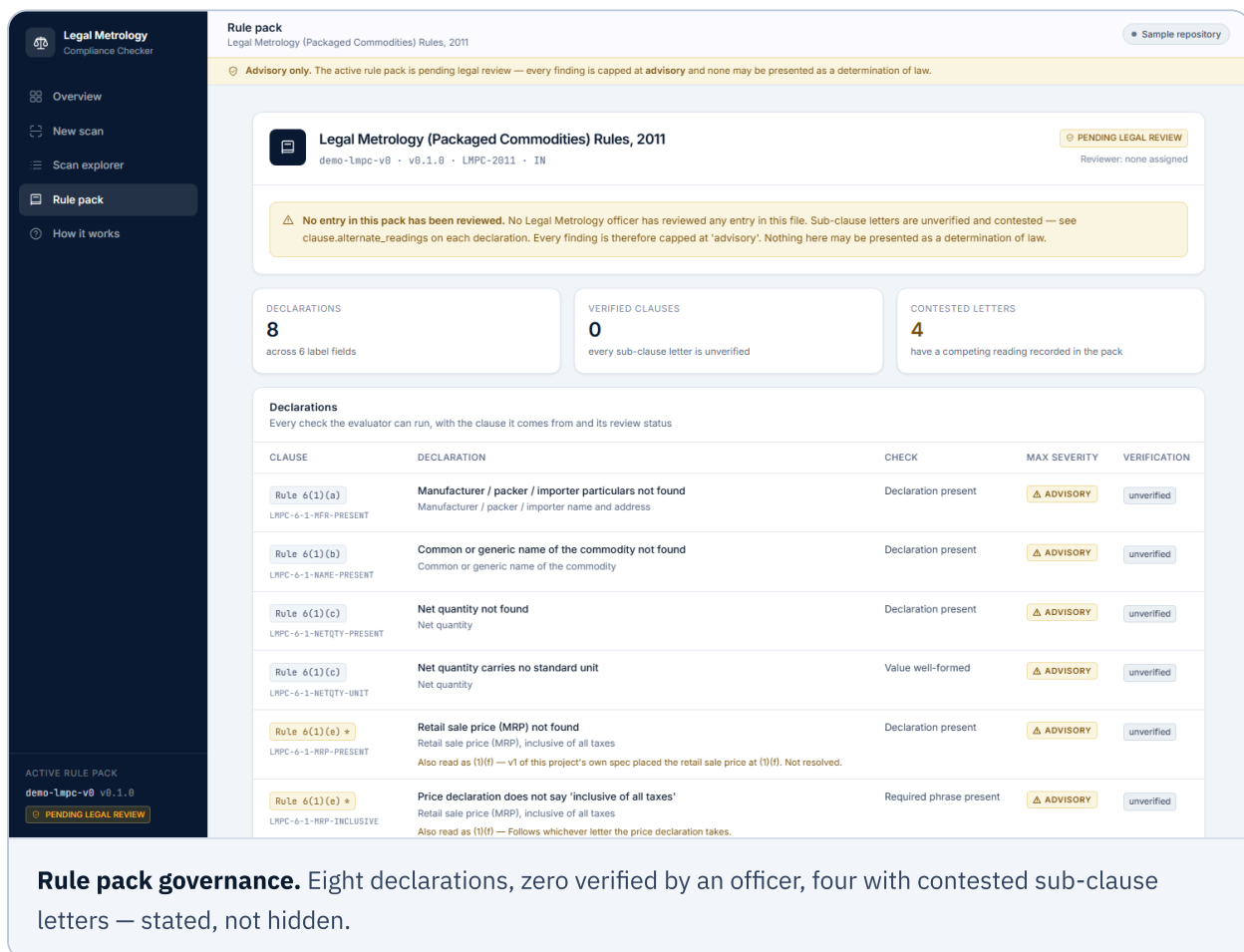
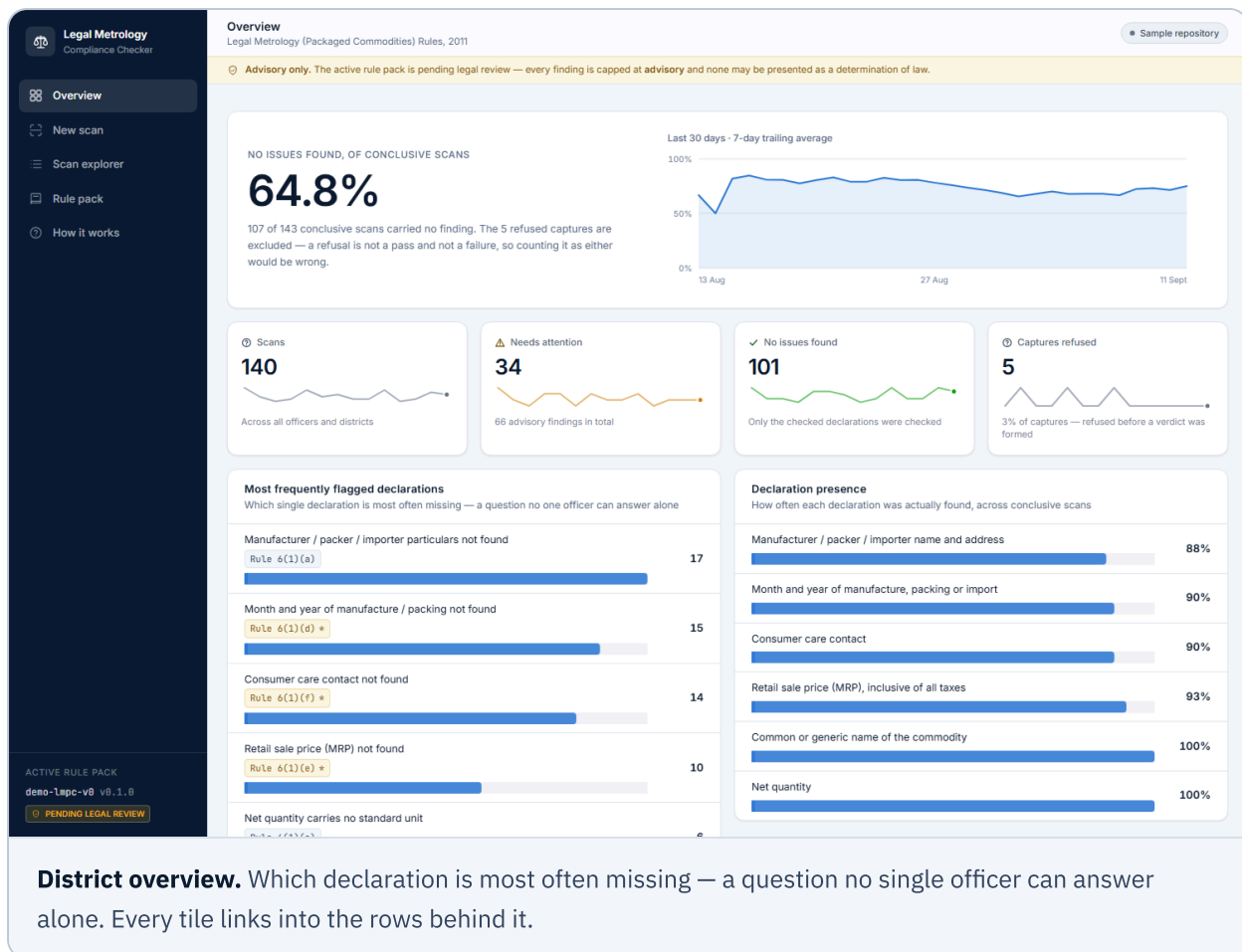
present

LMPC-6-1-NFR-PRESENT

Manufacturer / packer / importer name and address

Every package must carry the name and complete address of the manufacturer, or of the packer or importer where the manufacturer is not the packer.

Compliance report. Each located declaration is outlined on the label; each check shows its clause, and an asterisk marks a sub-clause letter that is disputed rather than settled. The banner caps every finding at advisory until legal review.



Legal Metrology

Compliance Checker

Overview

New scan

Scan explorer

Rule pack

How it works

ACTIVE RULE PACK

demo-lmpc-v9 v9.1.0

PENDING LEGAL REVIEW

New scan

Legal Metrology (Packaged Commodities) Rules, 2011

Sample repository

Advisory only. The active rule pack is pending legal review — every finding is capped at advisory and none may be presented as a determination of law.

Capture the declaration panel

The rear or side panel carrying the mandatory declarations — not the front of the pack

Drop a label image, or click to browse

JPEG or PNG, up to 12 MB

Use camera

No packet to hand? Try a specimen panel — it runs through the same pipeline.

Sunrise Gold

Product Name (Cereal Biscuits)

MANUFACTURED BY: M/s. Sunrises Food Products Pvt. Ltd., Bangalore, India

NET QUANTITY: 250 g

Biscuits

Full rear panel

Refined Sunflower Oil

Product Name (Edible Oil)

MANUFACTURED BY: M/s. Sunrises Food Products Pvt. Ltd., Bangalore, India

NET QUANTITY: 1000

Edible oil

Full rear panel

Shelf photo

Taken from a distance

Analyse label

Recognition engine

Only the reading changes. The rules, citations and verdict are identical either way.

Tesseract (on this device)

Works with the network unplugged. Weak on creased, glossy or angled packaging — it matches glyph shapes and has no language understanding to fall back on.

Google Gemini (cloud)

Reads a damaged label the way a person does, using context. Requires a key and a network connection, and the image is sent to Google.

Pipeline

What runs, in order, once you press analyse

1 Reading text

Tesseract, in this browser — no image leaves this machine

2 Extracting declarations

Anchors first, then geometry, then shape alone

3 Admitting the frame and evaluating

Deterministic — no learned parameters

What the capture is checked for

Print size, as a fraction of frame height

$\geq 1.2\%$

Panel coverage, as a fraction of frame area

$\geq 2.6\%$

Text flush against the frame border

$\leq 34\%$

Sharpness is not checked — blur and glare are unmeasured. Both need raw pixel access this build does not have. A capture can pass every check above and still be too blurry to trust. This is stated rather than hidden, and no number is produced for it.

Capture & engine choice.

The recognition engine is selectable at run time — the adapter seam, made visible.

Legal Metrology

Compliance Checker

Overview

New scan

Scan explorer

Rule pack

How it works

ACTIVE RULE PACK

deno-lmpc-v8 v0.1.0

PENDING LEGAL REVIEW

How it works

Legal Metrology (Packaged Commodities) Rules, 2011

Sample repository

Advisory only. The active rule pack is pending legal review — every finding is capped at advisory and none may be presented as a determination of law.

A compliance tool that can be argued with

An enforcement officer checks perhaps eighty packages a day, against 8 mandatory declarations under the Legal Metrology (Packaged Commodities) Rules, 2011. The hard part was never spotting text. It is being able to defend the finding afterwards — which clause, which reading of it, and what on the package supports it.

So this is built the other way round from a classifier. Recognition may be statistical. The step that turns what was read into a flag is a **pure function** with no learned parameters, and every flag it emits carries its statute, rule, sub-clause, rule-pack version and the image region behind it.

The pipeline

Four stages, in the order they actually run

1

Recognise

Text and per-word geometry, on this device

→

2

Extract

Anchor, then geometry, then shape alone

→

3

Admit

Is the picture good enough to judge at all?

→

4

Evaluate

Pure function of facts, geometry and the pack

Admission runs third, not first. The plan wanted the picture judged before any recognition, to save the work. It cannot be: the only quality signals available without raw pixel access are made of text geometry, which does not exist until recognition has run. So the correctness half is achieved — no verdict is offered on a frame that fails — and the saving is not. That deviation is recorded rather than glossed.

CLAUSE NUMBERS IN CODE

0

All 8 live in ruLepack/*.json

VERIFIED BY AN OFFICER

0

so every finding is capped at advisory, in data

CONTESTED SUB-CLAUSES

4

shown with both readings, not resolved

Five decisions that shape everything else

These are the tiebreakers when two designs both look reasonable

P1

The decision layer is deterministic

Recognition can be probabilistic. Turning facts into a flag cannot. The evaluator is a pure function of (fields, geometry, category, rule pack) — the same inputs always give the same verdict, and a disagreement is reproducible rather than a matter of opinion. The same 8 checks run in this browser and on the phone, from one shared implementation.

P6

Rules are data; the code is an interpreter

No statutory fact — clause number, threshold, unit table, lexicon — appears in any source file. Amending the tool for a change in the Rules is a change to a JSON file.

Design principles, on screen.

The determinism boundary and the honesty limits are part of the product, not just the documentation.

Simplifying metrology compliance into a clear, three-party benefit network

The Government — "defensible in minutes"

- Cuts label cross-checking from minutes to seconds per product.
- Produces an inspection note carrying every citation and the rule-pack version it was decided under — the paperwork is the output, not an afterthought.
- Cross-references scans to surface repeat offenders across districts.
- A rule change reaches every officer as a data update.

The Businesses — "zero label mistakes"

- Same pipeline usable as pre-print label QA.
- Prevents fines and recalls by catching a missing declaration before the print run.
- A disputed finding can be examined: the clause, the region and the raw text are all on the notice.

The Consumers — "honest pricing & safety"

- Mandatory declarations — quantity, price inclusive of taxes, date, consumer care — are actually present and legible.
- Enforcement extends to e-commerce listings, not only physical shops. **PLANNED**

REGULATORY CODES & COMPLIANCE FRAMEWORK

- **Legal Metrology (Packaged Commodities) Rules, 2011** — framework governing standard units, packaging and consumer disclosures. Rule 6(1) universal declarations are implemented.
- **Rule 7** — letter and numeral size on the principal display panel; letters minimum 1 mm (2 mm embossed), numeral height graduated 1–6 mm by net quantity.
- **Rule 6(11) — Unit Sale Price**, mandatory since January 2024. Exempt where the package is ≤ 100 cm², MRP $\leq ₹35$, or MRP equals USP.
- **Standards of Weights and Measures Act, 1976** — foundation defining legal units.
- **Open question, stated deliberately.** The sub-clause letters under Rule 6(1) are not settled between readings. Rather than pick one, the pack records both and every affected finding shows the alternate. Nothing is presented as settled law before an officer confirms it.

ENGINEERING & RESEARCH BASIS

- **Deterministic rule engine over layout transformers.** Document-layout models (LayoutLMv3 and similar) were evaluated and rejected for the decision layer: an enforcement finding must be explainable and reproducible, with no training data or GPU dependency.
- **Recognition engines compared by measurement, not preference.** Tesseract page-segmentation modes, pre-recognition upscaling and contrast normalisation were benchmarked on a deliberately degraded label; contrast normalisation was found to *harm* the read and was dropped. PaddleOCR was integrated and removed — its browser build ships only a Chinese recognition model, which emits no word spaces and so defeats anchor-based extraction entirely.
- **Curved and wraparound labels** are handled by repositioning under live checklist feedback rather than a separate unwarping stage.
- **Bhashini Translation API (MeitY)** — referenced for sovereign-grade multilingual coverage beyond Devanagari. **PLANNED**

Worth a sentence out loud. The PaddleOCR result and the contrast-normalisation result are both negative findings. Presenting them is a strength: it shows the engineering was measured rather than assumed, which is exactly the disposition a government-facing panel is looking for.

Prepared for SIH 2026 · PS 26034 · Team Blitz. Every figure and screenshot on this page comes from the running system. Items marked **PLANNED** are specified and scoped, not implemented — keep that distinction in the room.